# fn log_mean_hi

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Proves soundness of `fn log_mean_hi` in `mod.rs`.

## 1 Hoare Triple

### Precondition

`mean` $> 0$ and `mean` is finite.

### Pseudocode

```
1 def log_mean_hi(mean: float):
2     if not (mean > 0.0 and mean.is_finite()):  #
3         raise "mean must be positive and finite"
4     mean_minus_one_hi = mean.inf_sub(1.0)  #
5     return mean_minus_one_hi.inf_ln_1p()  #
```

### Postcondition

**Theorem 1.1.** For any setting of the input parameter `mean` such that the given precondition holds, `log_mean_hi` either returns `Err(e)`, or returns `Ok(out)` where

$$\texttt{out} \geq \log(\texttt{mean}).$$

## 2 Proof

Assume the precondition is met.

**Lemma 2.1.** `log_mean_hi` only returns `Err(e)` if the guard on line 2 fails or if a later fallible helper call returns `Err(e)`.

*Proof.* The function explicitly returns an error when the guard on line 2 fails. Under the precondition, this branch is not taken. The only remaining fallible operations are the calls on lines 4 and 5. □

**Lemma 2.2.** If the outcome of `log_mean_hi` is `Ok(out)`, then

$$\texttt{out} \geq \log(\texttt{mean}).$$

*Proof.* By the proof definition of `inf_sub`,

$$\texttt{mean_minus_one_hi} \geq \texttt{mean} - 1.$$

Since `mean` $> 0$, both arguments to $\ln(1 + \cdot)$ lie in $(-1, \infty)$. By the proof definition of `inf_ln_1p`,

$$\texttt{out} = \texttt{mean_minus_one_hi.inf_ln_1p()} \geq \ln((\texttt{mean} - 1) + 1) = \log(\texttt{mean}).$$

□

*Proof of Theorem 1.1.* Holds by Lemma 2.1 and Lemma 2.2. □